Response to Reviewer FBhK

BioCoLoop, submission 5046 | Revision dated 11 November 2026. Prepared for author review; not posted to OpenReview.

Thank you for identifying two questions that need separate evidence: whether history improves search at fixed access and budget, and whether development-selected changes generalize. We have added the requested random/direct controls, a fresh three-seed history-representation experiment, and a direct diagnosis of the validation failures. The original results remain visible in consolidated tables.

1. Same ten laboratories, configuration library and search budget (Weakness 1; Q1)

Appendix E.1-E.2 reports the original matched-access comparison of random search without replacement, GP-EI, and direct/history-guided Qwen and Luna using the same ten training laboratories, ten development panels, twelve configurations, initialization and checkpoint-selection rule. Each search receives six proposal slots beyond D00, with 100 rounds per fit and a 700-round cap. Random search receives the loop's search budget, not the fixed model's single-fit budget.

We verified all 60 task/configuration fits. Enumerating all 462 six-design subsets gives the exact expected outcome of random search conditional on this cache, with selection based only on development evidence. This is not 462 independent training repetitions. Shared-first-proposal and realized-fit-count controls additionally account for failed LLM proposals; for example, four-valid-proposal random search reaches 26.30 on Tahoe versus Qwen's 25.37 under the same realized fit count.

The comparison identifies a positive case without implying a universal gain: Norman/Luna history reaches 23.11 Top-1, versus 22.30 for direct/GP-EI and 22.66 for random search. DTI favors direct Luna/GP-EI (94.60 versus history's 94.27); proteomics ties throughout. On VCC, direct/history and GP-EI retain the fixed reference at 31.50, while random search averages 27.53. We report all five endpoints and all controls.

The main access-scenario table answers a different question: what a laboratory gains by participating in collaborative research instead of developing a model with only its own measurements. Broader access adds training data, development evidence and local computation. We now distinguish this workflow comparison explicitly from the same-access search table; it is not evidence for history alone.

2. Fresh history-representation experiment

Appendix E.6 adds 48 newly trained design/seed fits and new paired Qwen2.5-7B-Instruct search runs. Four study-cell contexts from Replogle and Nadig provide 1,208 training genes, 414 development genes and 406 held-out genes, with gene-disjoint partitions. This is a separate mechanism benchmark, not a replacement score for the original cell endpoints.

Direct, raw-history and structured-history search share the first proposal and six-request budget. Raw and structured history observe the same queried development facts; structured cards explicitly organize configuration changes, reference comparisons and expected-versus-observed outcomes. Direct, structured history, random search and GP-EI each use seven full fits, 16,940 optimizer steps and 532,840 training-example exposures per seed. We report additional LLM costs separately. Raw history has three invalid slots across the three runs, which remain charged rather than silently replaced.

Mean held-out Top-1 is 22.625 for direct, 22.625 for raw history, 22.888 for structured history, 22.668 for random search and 23.152 for GP-EI. Structured versus direct yields two gains and one tie, averaging +0.263 points; its paired gene-cluster 95% interval is [-1.022, 1.568]. Thus the revised representation changes candidate selection and has a positive point estimate, but does not establish stable superiority over simpler search. These complete controls define the scope of our history claim.

The separate 10.84-point Norman result compares training allocation with uniform fitting, not LLM history. New promotion controls show that random promotion reaches 22.63 versus 22.32 for the trajectory policy. We retain that comparison and do not use it as evidence of a unique history mechanism.

3. Wider recipe choices at the same candidate budget

Appendix E.9 adds 216 fresh fits over 36 recipes and three head-training/controller seeds on DTI and proteomics. Recipes vary the fitted classifier or 32/128-wide residual, BCE versus BCE plus ranking loss, learning rate and constant/cosine scheduling. Every method shares ten laboratories, the original fixed-default seed42 starting predictor, development data and six requests; every valid fit completes 20 rounds. This is a separate warm-start study, conditional on one common historical starting model per task, not three new end-to-end backbone fits. DTI freezes the original TAPB encoder; protein can update its complete observed-response head or learn a residual from all original protein/drug inputs.

Luna structured history reaches DTI AUROC 94.368 versus random's 94.316 (+0.053 points; conditional 95% interval [0.031, 0.077], positive in all three runs). On protein, direct reaches 39.017 AP versus random's 38.655, while structured/raw history reaches 38.572. Thus wider recipes supply above-random model outcomes on both tasks, but the primary history comparison does not win both: protein structured-minus-random is -0.083 points [-2.025, 2.234]. Both Qwen and Luna, fixed/direct/raw/structured/random and GP-EI at two offsets remain visible. Protein direct and the original GP-EI select identical predictors in each seed.

Random search exactly integrates all 1,623,160 six-design subsets; its reported result is a selected single-model expectation. All twelve task/seed/proposer selections are sealed before scoring, and saved predictions independently reproduce all 216 scores. Common historical fitting, feature extraction, the 110-fit development pilot and interrupted engineering calls are separate costs. These original test sets were already exposed, so this follow-up is retrospective rather than a new blind or biological-split confirmation. We retain the unsuccessful protein history contrast and the stronger conventional-search outcomes.

4. What explains the development-test divergence? (Weakness 2; Q2)

We reconstructed all 14 distinct checkpoints selected in the extended VCC/Tahoe searches, re-evaluated their 140 development panels, and independently reproduced all 80 archived prefix test scores. There is no aggregation mismatch in these cases: each VCC panel contains two development interventions and each Tahoe panel six, so equal-panel macro Top-1 is exactly the pooled intervention-macro score, even when query counts differ.

The development population is small: 20 VCC interventions and 60 Tahoe compounds. Qwen/VCC's +1.17 development points accompanies -11.89 test points; Tahoe's +5.00 development points represents three additional correct development compounds, while test performance decreases by 11.94 points. To test dependence on the option sets, we evaluated the same initial/final checkpoints on 20 paired development-roster redraws, without training or reselection. Their mean advantages become -1.34 and -1.33 points, positive in only 6/20 and 7/20 redraws. This is direct evidence of roster sensitivity, not proof that one particular cause explains all failures. Finite-sample selection instability is a known model-selection concern Cawley and Talbot, 2010.

The larger follow-up retains every development/test prefix for all methods and three seeds. All final search outcomes exceed their fully fitted D00 references, so the earlier large terminal decreases do not recur. A small interim reversal remains: in seed 62, raw and structured history select D05 at slot 2, improving development by 0.057 points but reducing test by 0.257; later selections recover. A context-specific reversal also remains on K562. Because the data, response panel and candidate library differ, we do not attribute this change to feedback formatting alone or claim that selection error is eliminated. The follow-up supplies a new gene-disjoint study, not repeated regroupings of the original VCC/Tahoe partitions.

5. Source-defined panel feedback and remaining selection instability

Appendix E.8 repairs and audits a separate implementation issue in the older source-defined protein/cell adapters. Ten target development panels were averaged inside one worker, so the outer worst-client field equalled the mean instead of the true worst panel. This did not change the primary selection score. A new adapter/reducer/projection version preserves explicit panel labels, primary/loss and dispersion, gives raw/structured cards identical observed facts, and keeps direct free of outcome feedback.

We froze the audit before new development inference and reconstructed every saved best checkpoint: two tasks, twelve designs and three seeds, or 72 checkpoints and 720 labelled

panel scores. Primary/loss reproduce exactly; mean and true worst-panel scores differ in all 72 cases. Eight new paired ten-round CPU canaries on D00/D07 preserve training losses, selected rounds, selected weights and final training states. These are engineering checks, not full-budget search repetitions or evidence of a new history gain.

Selection sensitivity remains. With even-indexed panels used to select among the saved checkpoints, cell seeds 61/62 gain 17.13/13.58 points on the proposal subset but lose 2.68/8.54 points on its odd-indexed complement; seed 63 improves both. We retain all six task/seed rows and all 252 five-panel subset diagnostics. Because these checkpoints were originally selected on all ten panels, the complements are descriptive, not independent confirmation. A new selection study therefore had to choose checkpoint rounds and candidates without consulting the confirmation panels.

We then ran this independent selection study (Appendix E.10, Table "Split-panel confirmation"). We retrained all twelve source-defined designs at seeds 61-63 on both tasks, 72 fresh fits that reproduce the original caches exactly, and saved a checkpoint at every evaluated round. In each direction, checkpoint rounds and all searches use only one five-panel half: fixed D00, exact random search, GP-EI, and Qwen/Luna direct, raw-history and structured-history search, each with six choices. Both directions were sealed before a logged gate first opened the other half; the test set was scored only afterwards.

The results locate the main risk in development selection itself. Averaged over search arms, scores fall from the selection half to the confirmation half by 20.78/9.81 points on cell and 4.61/53.39 on protein (A-to-B/B-to-A). On protein, the two halves favor early and late checkpoints respectively. On cell A-to-B, every search is below fixed D00 on the test set (at most 22.60 versus 30.82). In B-to-A, GP-EI and every Luna mode choose D05 in all seeds, raising confirmation from 12.93 to 21.45 and the test score from 28.36 to 28.67. History and direct search choose the same predictor in 8 (Qwen) or 9 (Luna) of 12 runs. Where they differ, Qwen history improves confirmation in all four remaining runs but the test score in only two. Luna history raises the test score in all three but lowers confirmation in two (raw) or three (structured). Qwen history also loses 26/16 of 72 slots to invalid proposals, against none for direct. We therefore do not claim a consistent history advantage over direct, random or GP-EI search. This is an independent development confirmation on previously inspected panels and tests, not a new biological split.

Robustness across new biological partitions remains unestablished.

6. What changed in the manuscript

Appendix E.7 reports 40 fresh fixed-D00 fits over ten laboratory identities and four endpoints, with scores independently reproduced. Laboratory 0 is the best on Norman and below the independent-laboratory mean on protein/VCC. This one-seed diagnostic does not rotate search controllers or address DTI.

Consolidated tables retain all original short/extended outcomes and trajectory diagnostics. All six or 24 slots were executed, so retrospective target crossings are not realized compute savings. Sections 4.7-4.8 and Appendix E distinguish collaborative access, history, allocation and validation evidence.

Response to Reviewer AfJB

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Thank you for identifying the relevant prior work and the need to separate model-development benefits from additional laboratory access. We have revised both the positioning and the experimental comparison rather than treating the original main table as evidence for the history mechanism.

1. Relation to existing collaborative research systems

The revised related-work section discusses all three systems explicitly. The Virtual Biotech coordinates specialist agents around biomedical evidence for therapeutic decisions. AutoScientists maintains shared experimental state, including unsuccessful directions, and coordinates subsequent experimentation. Helmsman is directly relevant to automated federated design: its planning, coding and simulation-based diagnosis already connect LLM agents to federated system construction and refinement. We have updated its citation to the ICLR 2026 publication.

BioCoLoop's contribution is the biological model-development interface and its controlled evaluation. Task adapters connect interaction, observed-response proteomic and perturbation predictors to the same local-fitting, development-evaluation and proposal interfaces. The coordinator distinguishes training-example-weighted parameter updates from equal-panel development evidence. Source-specific heads allow studies with different response objectives to contribute to a shared representation while the target development population remains explicit. The evaluation then separates laboratory participation, proposal-history conditioning and training allocation.

This is a narrower and more concrete contribution than inventing shared research memory or automated federated design. We specify the original four-factor finite space and each expanded follow-up library directly, and do not portray it as open-ended architecture or source-code discovery. The abstract, introduction and conclusion have been rewritten around the implemented framework and the evidence for each decision.

2. Does the history mechanism itself improve search?

We added random search without replacement and fixed GP-EI Bayesian optimization under the same ten-laboratory access as direct and history-guided LLM search. All methods use the same twelve configurations, development panels, initialization and six-proposal cap. Appendix E.1-E.2 retains all five original endpoints, exact random-search variability, shared-first-proposal and realized-fit-count controls. Section 4.7 now leads with the three-seed follow-ups across the task families, with complete results in E.6 and E.9.

The result is not a general history advantage. Qwen direct and history select equally scoring models on all five endpoints. Luna history improves Norman from 22.30 to 23.11 Top-1, versus a random expectation of 22.66, but decreases DTI from 94.60 to 94.27 AUROC. GP-EI matches direct Luna on DTI. Proteomics retains the starting configuration, and the VCC score attained by the LLM searches is already attained by the fixed configuration. These controls directly address the alternative explanation that conventional configuration search is sufficient in much of this small design space.

The comparison reuses verified complete candidate fits, with development-only selections sealed before additional scoring. It is a retrospective one-training-seed-per-endpoint study, not new multi-seed evidence. Three-seed source-defined laboratory controls are also reported separately: Qwen history is below direct on both proteomics and cell prediction, and the additional Luna feedback experiment yields only a small protein change and cell ties. We retain these results rather than extending the combined-system gain to unsupported component claims.

Appendix E.6 adds a separate three-seed mechanism study with 48 fresh fits over 16 designs, using four study-cell contexts from Replogle and Nadig and gene-disjoint train/development/test partitions. Direct and structured history each receive six requests, 16,940 optimizer steps and 532,840 training-example exposures per search. Raw and structured history see the same queried development facts. Held-out Top-1 is 22.625 for

direct/raw, 22.888 for structured, 22.668 for random and 23.152 for GP-EI. Structured-minus-direct is +0.263 points with paired gene-cluster 95% interval [-1.022, 1.568], conditional on the fitted, selected models. This changes search decisions but does not establish a stable independent history advantage or replace the original endpoints.

We also repaired a concrete feedback-transport omission in a new implementation version (Appendix E.8): the original source-defined adapters averaged ten panels before returning one worker result, losing the true worst-panel diagnostic while preserving the main score. Reconstructing all 72 source-defined checkpoints reproduces primary/loss exactly and restores 720 labelled panel scores. Eight paired ten-round CPU canaries preserve selected weights and training trajectories. These checks establish faithful transport, not an improvement in history-guided search. A new sealed split-panel study (Appendix E.10) then chooses rounds and designs on five panels and confirms them on the other five. History and direct search choose the same predictor in 17 of 24 proposer/task/direction/seed runs. The remaining differences are mixed in sign, and Qwen history also loses more slots to invalid proposals. We therefore do not claim that history itself improves search; scores fall sharply between selection and confirmation panels for every policy.

3. What the revised empirical contribution supports

The original main table is now explicitly an access-scenario comparison, with a Labs column. The new same-access table addresses search quality. Re-evaluation and new development-roster diagnostics identify validation sensitivities associated with poor held-out generalization.

Appendix E.7 additionally reports 40 fresh fixed-design CPU fits across all ten single-laboratory identities on four endpoints. Laboratory 0 is the highest-scoring independent laboratory on Norman; identity variation therefore matters. This one-training-seed sensitivity does not rotate every controller, provide a DTI rotation, or equalize one- versus ten-laboratory optimizer steps.

We also completed 360 fresh fits to compare the allocation policy with uniform, ordinary score-only, learning-rate-stratified score-only and random promotion at identical training/evaluation counts. The original Norman gain over uniform is reproduced, but score-only promotion recovers most of it (21.70 versus 22.32), and the exact random-promotion expectation is 22.63. The trajectory policy improves Tahoe relative to both alternatives, while random is better on VCC. We report these mixed controls in Section 4.4 and Appendix E.5 rather than claiming a generally superior new allocation algorithm. The revised contribution is the implemented biological interface and its controlled empirical assessment, with data-access, search and allocation effects distinguished.

4. Broader recipes and the remaining attribution boundary

The additional warm-start study in Appendix E.9 trains 216 new task/recipe/seed fits over 36 recipes, expanding head architecture, objective and scheduling choices beyond the original small menu. All search policies share ten laboratories, the same starting model per task, six requests and 20 rounds per valid fit. New training/controller seeds are conditional on the common old seed42 start, not independent end-to-end backbone seeds. Both proposer models and conventional GP-EI at two offsets are reported.

Luna history reaches 94.368 DTI AUROC versus random's 94.316; the paired conditional difference is +0.053 points [0.031, 0.077]. Protein direct reaches 39.017 AP versus random's 38.655, but structured history is 38.572 and its contrast is -0.083 [-2.025, 2.234]. The stronger model-development outcomes therefore do not establish that history is necessary: raw and structured agree for Luna, and GP-EI matches protein direct with identical selected checkpoints. All 216 prediction scores are independently reproduced after the twelve development selections are sealed. These previously exposed tests support a retrospective recipe comparison, not a new generalization claim. Our contribution remains the implemented biological interface and its controlled assessment, rather than a claim to have invented the recipe components or a generally superior history optimizer.

Response to Reviewer cR5g

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Thank you for the concrete concerns about conventional tuning, laboratory heterogeneity and the interpretation of the baseline comparison. We have added the missing controls and reorganized the evidence around those questions.

1. Conventional hyperparameter optimization

Appendix E.1-E.2 adds the original same-access table comparing random search without replacement, fixed GP-EI Bayesian optimization, direct LLM search and history-guided LLM search. All use ten laboratories, identical development panels, the same twelve-design library and a six-proposal cap after the initial configuration. Exact enumeration of 462 candidate subsets reports random search's expected outcome and search-policy variability; it does not count these subsets as independent training seeds.

For example, DTI AUROC is 94.60 for GP-EI, 94.45 in expectation for random search and 94.27 for history-guided search. On Norman, Luna history reaches 23.11 Top-1, versus 22.30 for GP-EI/direct and 22.66 for random search. Other endpoints include ties and cases where the fixed configuration is already competitive. Thus the new evidence supports task-dependent search behavior, not general superiority to conventional tuning. We also report controls matching the actual number of valid fits when an LLM proposal fails.

A separate three-seed, gene-disjoint study adds 48 fresh fits over 16 designs and four Replogle/Nadig study-cell contexts (Appendix E.6). With six requests and matched fitting counts, structured history reaches 22.888 Top-1, direct/raw 22.625, random 22.668 and GP-EI 23.152. Structured-minus-direct is +0.263 points, with a conditional paired gene-cluster 95% interval [-1.022, 1.568]. We report every control and do not claim that this study establishes superiority to ordinary tuning.

The new sealed split-panel study (Appendix E.10) repeats this comparison when checkpoint rounds and designs are chosen on five development panels and confirmed on the other five. GP-EI gives the best or tied-best test score in three of four task/direction settings, and exact random search lies within 1.2 points of the best arm on protein. The only LLM mode with a higher average than GP-EI is Qwen structured on cell A-to-B (22.13 versus 20.31), where every search arm remains below fixed D00. Conventional search is therefore a strong baseline in this design space, and we present it as such.

2. Laboratories defined by different studies

The main ten-client experiment partitions one source dataset; it is not ten independently operated institutions. We now make this explicit and separate it from the source-defined laboratory experiments.

The proteomic experiment treats PTPC, Lin, Ruprecht and decryptE as four laboratories. Their different response objectives use source-specific heads and a shared representation; the same target development panels select checkpoints. In the fixed-design transfer comparison, adding the three external studies increases target AP from 34.41 to 37.53.

The cell control compares four same-source VCC groups with VCC, Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562 as four source-defined laboratories. Each arm has four laboratories and 60 training interventions, with the same target anchor. The external-source arm scores 28.82 Top-1 versus 30.70 for the same-source arm. Underlying cell counts and gene panels remain different and are documented. DTI likewise reports each auxiliary source: Davis/Human improve the fixed target model, while BioSNAP and the combined-source setting do not.

These experiments test real differences between released studies, although execution is simulated on one host. We do not present heterogeneous data as automatically beneficial. Appendix E also reports the completed three-seed direct/history controls within the source-defined protein and cell settings, including their negative results.

Appendix E.8 separately audits a repaired feedback payload for these source-defined adapters. Originally, ten target panels were averaged inside one returning worker, so the outer worst-client field incorrectly described that mean rather than the true worst panel.

The repair retains panel labels, scores and dispersion without changing primary/loss selection. Reconstructing 72 saved checkpoints yields identical primary/loss and restores 720 panel scores; eight paired ten-round CPU canaries preserve training trajectories and selected weights. The published source-defined search results remain unchanged, and a search benefit from this repair has not yet been measured.

3. Why direct search can be worse than the fixed model

“Direct” denotes feedback-free LLM configuration search, not training one model once. Every method ultimately deploys one selected predictor; ten laboratories are data holders contributing to fitting, not a ten-model ensemble. Both direct and history search retain the model with the best development score, whereas “fixed” is a single predefined configuration. Retaining a better development model therefore need not retain a better test model.

We investigated this failure rather than changing the reported test selection. Reconstructed VCC/Tahoe predictions show that equal-laboratory and pooled-intervention macro aggregation agree exactly. Small gains on 20 VCC or 60 Tahoe development interventions can accompany much larger test decreases. Re-evaluation under 20 new paired development-option rosters further shows that Qwen/VCC and Tahoe's original development advantages are not stable to distractor/option-order changes. No models were retrained or reselected in this diagnostic.

The new source-defined panel diagnostic also retains mixed outcomes. Selecting among saved checkpoints on five fixed panels improves cell seeds 61/62 by 17.13/13.58 points on those panels but decreases the complementary five by 2.68/8.54 points. Those checkpoints were already selected using all ten panels, so we added an independent test (Appendix E.10): 72 fresh fits with per-round checkpoints, rounds and designs chosen on one half, and both directions sealed before the other half was opened. Selection-to-confirmation drops average 4.6-53.4 points across task/direction settings, and on cell A-to-B every search arm falls below fixed D00 on the test set (at most 22.60 versus 30.82). This confirms that small development panels, rather than the feedback format, limit direct and history search here. It does not create a new biological split. The larger gene-disjoint study improves all terminal searches over their fitted D00 references but still includes an interim and a context-specific development/test reversal; validation error is not eliminated.

4. Scope and presentation of the contribution

The original access comparison now has an explicit Labs column and is no longer used to attribute the benefit of additional data to the LLM loop. The same-access search table addresses that attribution directly. We describe the method as a common biological model-development interface with four editable configuration factors, distinct from open-ended model invention. Related work now discusses Virtual Biotech, AutoScientists and Helmsman. A separate 360-fit allocation comparison adds ordinary score-only and random promotion: it reproduces the original benefit over uniform allocation but shows that much of it is also attained by simpler policies. The revised conclusion states these results separately from collaborative fitting and proposal history.

We additionally fitted fixed D00 independently at every laboratory identity on four endpoints: 40 fresh CPU fits at one seed (Appendix E.7). Laboratory 0 is the best on Norman and below the independent-laboratory mean on protein/VCC. We therefore report the full identity distribution separately from the original multi-seed workflow comparison. Full controller rotations, the DTI rotation and one-laboratory training matched to ten-laboratory optimizer steps remain outstanding; equal communication rounds alone do not equalize computation.

5. Increased recipe freedom under matched access and request budgets

Appendix E.9 expands the library to 36 recipes, with 216 fresh fits over two tasks and three head-training/controller seeds. Besides learning rate, the search can choose the original fitted head or a 32/128-wide residual, ordinary BCE or BCE plus ranking loss, and constant/cosine scheduling. DTI adapts the classifier on the same frozen original TAPB encoder; protein can update the complete original observed-response predictor or learn a residual from all original inputs. This warm-start comparison is separate from the original end-to-end study. Each policy has the same ten laboratories, development rows, six requests and 20 rounds per valid fit; equal steps/exposures do not imply equal FLOPs across head choices.

Luna reaches 94.368 DTI AUROC in direct/raw/structured modes versus random's 94.316. Protein direct reaches 39.017 AP versus random's 38.655, whereas structured/raw history reaches 38.572. Random exactly integrates all 1,623,160 six-design subsets, and both original GP-EI and an additional zero-offset GP-EI control are retained. Original GP-EI matches protein direct using the same selected predictors. We therefore report improved model-performance options without recasting ordinary tuning as a unique history benefit. All twelve selections are sealed and all 216 scores independently verified; these historically exposed tests remain retrospective, not an independent new split.